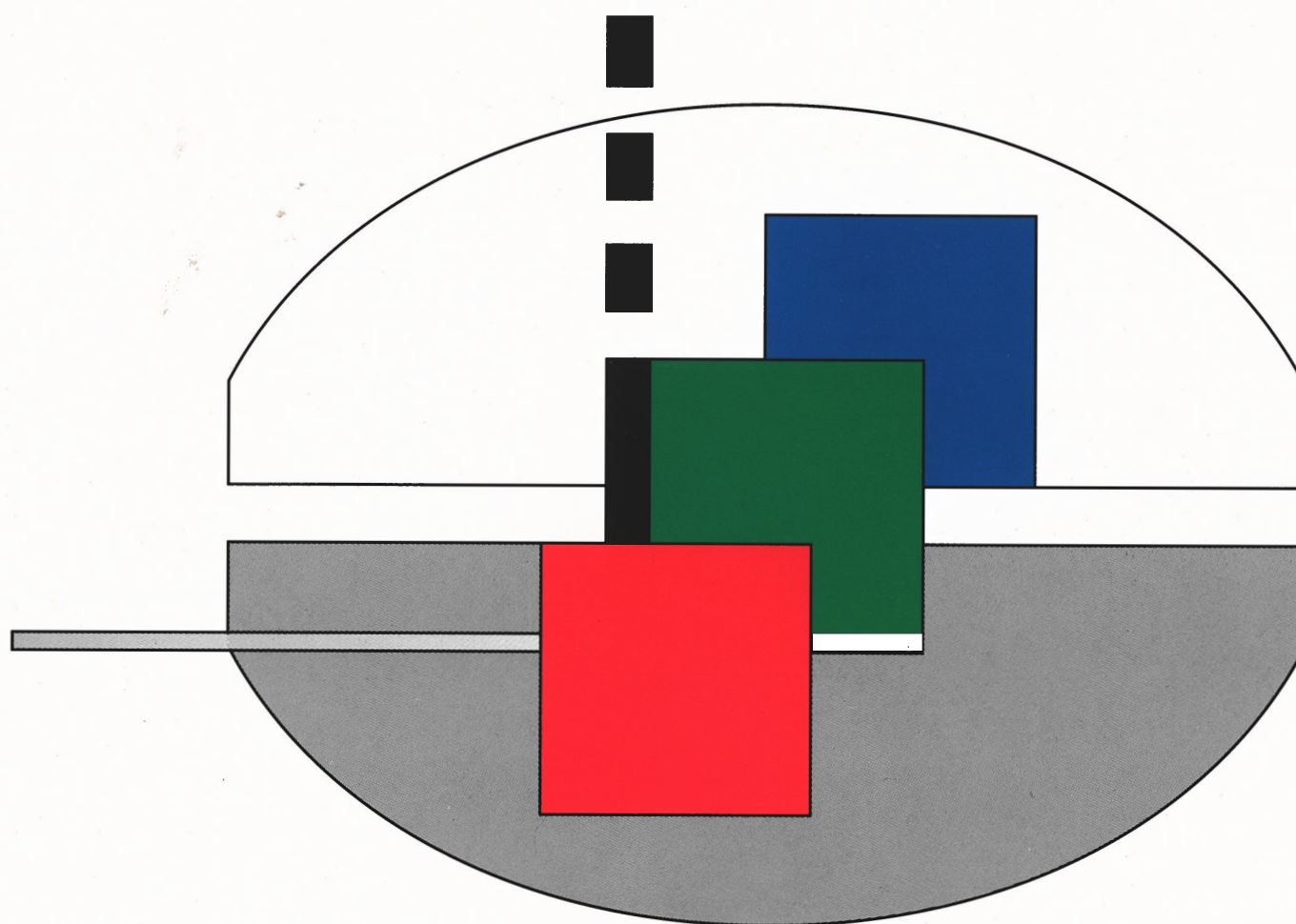


Instruction Manual



Philips TV Test Equipment



PHILIPS

Color pattern generator

PM 5644

Instruction manual



9499 493 07211
910701/01

Important

In correspondence concerning this instrument, please quote the full type and KU. number (serial number) as shown on the identification plate on the rear of the instrument.

Training

Courses of technical training on this, and other current PTV equipment are available either on a pre-planned basis, or to suit individual requirements.

These courses are held in English.

For details of location, time and duration of pre-planned courses, or the possibilities that exist for individually planned courses, please complete the form on the next page and return it to the Philips Customer Support Organization in your country.

Fault analysis

In the event of instrument failure, please complete the "Fault Analysis report" situated in Chapter - Service hints of this manual. After completion, please send it to the Philips Customer Support Organization in your country.

Text en logo configuratie

Klok en datum verwijderen

- Zet het apparaat uit
- Verwijder boven- en frontpaneel
- Verzet jumper van PP5 (neutrale stand) naar PP2 (speciale functies)
- Zet het apparaat weer aan.
- Druk nu eerst op toets S10, de LED's van de insteltoetsen lichten nu op.
- Druk dan op S8. Door enkele keren op S1 te drukken kun je nu de datum en/of tijd verwijderen.
- Zet het apparaat weer uit en verzet de jumper van PP2 nu weer terug op PP5

Tekst inzetten

- Schakel het apparaat aan
- Druk op S10, de LED's van de toetsen lichten op
- Door nu op S9 te drukken verschijnt er in de bovenste balk een cursor
- Met S2 kun je de goede letter kiezen
- Met S3 kun je de cursor naar de volgende letter brengen
- Als de tekstbalk klaar is druk dan op S5
- Met knop S9 kun je nu door de laatste drie punten te herhalen de volgende balk invullen
- Druk als je de tweede balk ingevuld hebt eerst op S9 om de tekst vast te zetten, en hierna op S10, en de tekst staat in de testcirkel.

	S20		S	1	2	3	4	5		8	9	10	
	☐			☐	☐	☐	☐	☐		☐	☐	☐	

PM 8546

(print achter in apparaat)

PP

1	2	3	4	5
☐	☐	☐	☐	☐

☐ 7

☐ 8

Request for training

Recent advances in equipment design and our commitment to the use of leading edge technology has placed increasing demands on servicing personnel world-wide. While we make every effort to ensure that our manuals reflect these advances (at the same time remaining service oriented), the need for effective and regular training has become more and more necessary.

Tailor-made courses of instruction on current PTV instruments are therefore being made available to complement the information contained in our manuals. Participation will inevitably lead to reduced down-time and repair costs.

If you are interested in our training program or just require information, please complete the information sheet below, and send it to the Philips Customer Support Organization in your country who will contact you and provide more detailed information for your consideration.

We require information only : ☐

Company name : _____

Address : _____

We like training on the following instruments :

PM _____ PM _____ PM _____ PM _____

PM _____ PM _____ PM _____ PM _____

Type of course required :

Systems/Applications ☐

Servicing/Faultfinding ☐

Other (please specify below) ☐

Number of participants (max. 12): ☐

Time available (days): ☐

Location :

On my own premises ☐

At Philips Customer Support ☐

In Copenhagen ☐

Please send this request to the Philips Customer Support Organization in your country. Thank you.

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GUARANTEE STATEMENT

The Philips guarantee is in addition to all rights which buyer may have against his supplier under the sales agreement the buyer and the supplier and according to local legislation.

Philips guarantees this product to be free from defects in material and workmanship under normal use and service for a period of one (1) year from the date of shipment. This guarantee does not cover possible required re-calibration and/or standard maintenance actions. This guarantee extends only to the original purchaser and does not apply to fuses, batteries or to any product or part thereof that has been misused, altered or has been subjected to abnormal conditions of operation and handling.

Philips - supplied software is guaranteed to be properly recorded on non-defective media. We will replace improperly recorded media without charge for 90 days after shipment upon receipt of the software. Our software is not guaranteed to be error free.

Philips' obligation under this guarantee is limited to have repaired or replace a product that is returned to an authorized Philips Service Centre within the guarantee period, provided that Philips determines that the product is defective and that the failure has not been caused by misuse, alteration or abnormal operation.

Guarantee service for products installed by Philips will be performed at the Buyer's facility at no charge within Philips' service travel area; outside this area guarantee service will be performed at the Buyer's facility only upon Philips' prior agreement and the Buyer shall pay Philips round trip travel expenses.

If a failure occurs, send the product, freight prepaid to the Service Centre designated by Philips with a description of the difficulty. At Philips' option, repairs will be made or the product will be replaced. Philips shall return the product, F.O.B. Repair Centre, transportation prepaid, unless the product is to be returned to another country, in which case the Buyer shall pay all shipping charges, duties and taxes. Philips assumes NO risk for damage in transit.

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Some countries or states do not allow the foregoing limitations. Other rights may also vary.

GENERAL INFORMATION

1. Safety

Read this chapter carefully before installation and use of the instrument.

1.1 Introduction

The instrument described in this manual is designed to be used by properly-trained personnel only. Adjustment, maintenance and repair of the exposed equipment shall be carried out only by qualified personnel who are aware of hazards involved.

1.2 Safety precautions

For the correct and safe use of the instrument, it is essential that both operating and servicing personnel follow generally accepted safety procedures in addition to the safety precautions specified in this manual. Specific warning and caution statements, where applicable, are found throughout this manual. Warning and caution statements and/or symbols are marked on the instrument where necessary.

1.3 Caution and warning statements

Caution

Used to indicate correct operation or maintenance in order to prevent damage to, or destruction of equipment or other property.

Warning

Used to indicate a potential hazard that requires correct procedures or practices in order to prevent personal injury.

1.4 Impaired safety protection

Whenever it is likely that safe operation is impaired, the instrument must be made in-operative and secured against unintended operation. The appropriate servicing authority must be informed. For example, safety is likely to be impaired if the instrument fails to perform the intended measurements or shows visible damage.

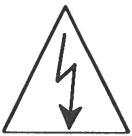
1.5 Electrostatic sensitive devices

All ICs and many other semi-conductors are susceptible to electrostatic discharges (ESD). Careless handling during repair can reduce lifetime drastically.

When repairing, make sure that you are connected to the same potential as the mass of the set via a wrist wrap with resistance. Keep components and tools also at this potential.



1.6 Symbols

Symbol:	Color:	Explanation:
	Red	High voltage terminal: a terminal at which a voltage, with respect to another terminal or parts exists or may be adjusted to 1000V or more. (High voltage $\geq 1000V$).
	Black/Yellow	Live part
	Black/Yellow	To preserve the instrument from damage the operator must refer to an explanation in the instruction manual.

2. Introduction and applications

2.1 General

The PM 5644 Color pattern generator produces a test pattern that allows assessment of all important parameters of a TV set at one glance at the picture tube itself. The generator comes as standard with the well known Philips test pattern and is in the PAL-B/G version also available with the German FuBK pattern. By means of the optional text and clock generator PM 8546 a clock that features time and date display as well as a logo combined with text for transmitter identification may be inserted into the test pattern.

– Resolution and bandwidth.

– Linearity.

Chrominance section:

– PAL delay line.

– Demodulators (synchron detectors).

– Matrix circuits.

– Bandwidth.

– Chrominance/luminance delay.

– Burst gating.

– Greyscale tracking.

– General color assessment.

– Combfilters

Miscellaneous:

– Reflections

– Intermodulation

– Focus

2.2 Applications

The generator is perfect for set makers that require a comprehensive test signal for their final inspections as well as for alignments of sets at the production line. The Philips test pattern has furthermore been adopted by broadcast authorities in a high number of countries worldwide as the test pattern to be transmitted outside programme hours. The pattern is also ideal for stand-by use in studios, OB-vans, e.t.c.

The generator is especially suitable for control and verification of the following parameters:

2. FuBK pattern

Deflection:

– Scanning linearity.

– Convergence.

– Pin-cushion correction.

Luminance section:

– Low frequency response.

– Transient response.

– Resolution and bandwidth.

– Linearity.

Chrominance section:

– PAL delay line.

– Demodulators (synchron detectors).

– Matrix circuits.

– Chrominance/luminance delay.

– Burst gating.

– Greyscale tracking.

– General color assessment.

Miscellaneous:

– Reflections

– Intermodulation

– Focus

1. PHILIPS pattern

Deflection:

– Picture size.

– Scanning linearity.

– Convergence.

– Pin-cushion correction.

Synchronization:

– Sync separator function.

– Interlacing.

Luminance section:

– Brilliance setting.

– Low frequency response.

– Transient response.

3. Technical data

3.1 Safety characteristics

This apparatus has been designed and tested in accordance with Safety Class I requirements of IEC publication 348 (Safety requirements for Electronic Measuring Apparatus), and has been supplied in a safe condition. This manual contains information and warnings which must be followed to ensure safe operation and to retain the apparatus in a safe condition.

3.2 Performance characteristics

Properties expressed in numerical values with stated tolerances are guaranteed by the Philips organization in your country. Specified non-tolerance numerical values indicate those that could be nominally expected as a mean of a range of identical instruments.

3.3 Versions

PM 5644D/00	PAL-D/K, Philips-pattern.
PM 5644G/00	PAL-B/G, Philips-pattern.
PM 5644G/10	625 lines GBR, Philips-pattern.
PM 5644G/20	625 lines YC _B CR, Philips-pattern.
PM 5644G/50	PAL-B/G, FuBK-pattern.
PM 5644G/70	625 lines YC _B CR, FuBK-pattern
PM 5644P/00	PAL-M, Philips-pattern.
PM 5644I/00	PAL-I, Philips-pattern.
PM 5644L/00	SECAM, Philips-pattern.
PM 5644M/00	NTSC-M, Philips-pattern.
PM 5644N/00	PAL-N, Philips-pattern.

3.3.1 Options

PM 8546G/00

Text and clock generator, 625-lines composite PAL versions.

PM 8546G/10

Text and clock generator, 625-lines component versions.

PM 8546M/00

Text and clock generator, 525-lines versions.

PM 8546L/00

Text and clock generator, SECAM version.

PM 9190

RS232 Remote Control Interface.

PM 9191

IEEE-488 Remote Control Interface.

3.4 Inputs

3.4.1 External sync input

The input signal for external synchronization may be either a black burst, a composite video signal or a mixed sync signal.

Connector:

2 x BNC.

Impedance:

75 Ω loop-through.

Return loss:

>40dB up to 7MHz.

Input level:

0.15V_{PP} to 4V_{PP} with up to 100% or 1V_{PP} hum.

In genlock mode the PM 5644 automatically switches to internal synchronization, if no input signal is available.

3.4.2 Time code/1Hz input (PM 8546)

Connector:

XLR female.

Impedance, LTC mode:

>10k Ω .

Impedance, 1Hz mode:

50 Ω .

Input level, LTC mode:

1 to 4V_{PP}.

Input level, 1Hz mode:

2 to 20V_{PP}

Duration of 1Hz pulse:

20 μ s to 0.5 s.

3.5 Outputs

3.5.1 Video outputs

COMPOSITE VERSIONS

Connector:

1 x BNC on front panel, 2 x BNC on rear panel.

Impedance:

75 Ω $\pm$ 0.5%.

Return loss:

>36dB up to 5MHz.

Amplitude accuracy:

better than 1%.

COMPONENT VERSIONS

Connector:

3 x BNC on rear panel.

Impedance:

75 Ω $\pm$ 0.5 %.

Return loss:

>36 dB up to 5 MHz.

Amplitude accuracy:

better than 1%.

Channel matching:

within $\pm$ 0.5 %.

Delay difference between channels:

within 5ns referred to output G(Y).

3.5.2 Black burst output

There is no black burst output in the SECAM version (PM 5644L/00).

Connector:

BNC.

Impedance:

75 Ω $\pm$ 1%.

Return loss:

>36dB up to 5MHz.

Residual subcarrier:

less than 3.5mV_{PP}.

Line to subcarrier phase:

Absolute phase : 0° $\pm$ 10°.

Jitter : typical $\pm$ 2°.

Drift : less than $\pm$ 2° per 10°C.

Sync amplitude:

300mV $\pm$ 2% (D, G, I, N and P versions)

286mV $\pm$ 2% (M version).

Sync timing:

see SYNC OUTPUT.

Burst amplitude:

300mV $\pm$ 3% (D, G, I, N and P versions)

286mV $\pm$ 3% (M version).

Burst width:

2.25 μ s $\pm$ 0.23 μ s (D, G, I and N versions)

2.5 μ s $\pm$ 0.15 μ s (M and P versions).

Burst position:

5.6 μ s $\pm$ 0.1 μ s (D, G, I and N versions)

5.3 μ s $\pm$ 0.15 μ s (M version)

5.8 μ s $\pm$ 0.1 μ s (P version)

Burst suppression:

lines 623-6, 310-318, 622-5 and 311-319

(D, G, I and N versions)

lines 1-9 and 264-272 (M version)

lines 522-7, 259-269, 523-8 and 260-270

(P version).

8-field reference, PAL versions:

line 7 field 1.

4-field reference, NTSC version:

line 11 field 1.

3.5.3 Sync output

Connector:

BNC.

Return loss:

30dB up to 4MHz.

Rise time:

200ns.

Amplitude:

4V ± 0.4 V or 2V ± 0.2 V (selectable).

Line sync duration:

4.7 μ s ± 0.1 μ s.

Equalizing pulses:

2.4 μ s ± 0.1 μ s.

Serration pulses:

4.7 μ s ± 0.1 μ s.

3.5.4 Audio output

Connector:

XLR male.

Frequency:

1kHz ± 0.1 Hz.

Distortion:

less than 0.1% THD.

Level:

0dBm/600 Ω (0.775mV/600 Ω).

3.6 Remote control (PM 9190-91)

Control of the internal clock generator and text insertion is possible with a remote control option.

3.6.1 RS-232/V24 remote control

PM 9190 RS-232 Interface.

Connector:

25 pin male D-connector in compliance with RS-232-C (EIA) and V24 (CCITT) standards.

3.6.2 IEEE-488 remote control

PM 9191 GPIB Interface.

Connector 24 pin female AMPHENOL in compliance with the IEEE-488 standard. Connection to IEC-625 standard is possible by using IEEE to IEC adaptor PM 9483/50.

3.7 Test pattern specifications

AMPLITUDE ACCURACY:

Luminance:

$\pm 1\%$.

Chrominance:

$\pm 2\%$.

Amplitude matching between channels:

$\pm 0.5\%$. (component versions)

Frequency response:

$\pm 1\%$ up to 5MHz, $\pm 2\%$ up to 5.5MHz.

Delay matching between channels:

within 5ns referred to G/Y. (component versions)

Pulse-to-bar ratio:

1:1 within 1%.

Field tilt:

less than 0.5%.

Line tilt:

less than 0.5%.

Sync amplitude:

nominal within $\pm 1\%$.

Sync front porch duration:

1.5 μ s ± 0.1 μ s.

Luminance rise time:

200 ns ± 10 ns for 625-line versions.

250 ns ± 10 ns for 525-line versions.

Spurious signals:

below -55dB referred to 1V for frequencies up to 5.5MHz, below -50dB referred to 1V for frequencies from 5.5MHz and upwards.

3.8 Pattern composition, Philips pattern

3.8.1 Signals inside the circle

From top to bottom:

Black rectangle on white background.

Width of rectangle:

11.3µs.

Black/white and white/black transition with needle pulse in the white part.

Width of pulse:

225ns ±5% (D, G, I and L versions)

280ns ±5ns (M, N and P versions).

250kHz squarewave signal.

Amplitude:

75% of white (same amplitude as the GBR components of the 75% color bar).

Color bar (without white and black).

D, G, L and N versions:

EBU (100/0/75/0).

I version:

BBC (100/0/100/25).

M and P version:

NTSC (77/7.5/77/7.5).

Centre cross.

Width of vertical lines:

225ns ±5% (D, G, I and L versions)

280ns ±5ns (M, N and P versions).

Structure of the horizontal line:

2 lines, one per field, sequence reversed compared to the lines of the background cross-hatch.

Multiburst.

Frequencies:

0.8, 1.8, 2.8, 3.8, 4.8, 5.63 MHz (D version)

0.8, 1.8, 2.8, 3.8, 4.8 MHz (G versions)

1.5, 2.5, 3.5, 4.0, 4.5, 5.25 MHz (I version)

0.8, 1.8, 2.8, 1.8, 0.8 MHz (L version)

0.5, 1.0, 2.0, 3.0, 4.0 MHz (M, P and N versions).

Amplitude:

100% of white level (71.4% in I version)

5-riser staircase.

White/black and black/white transitions.

Color steps.

Yellow/red/yellow.

Circle.

Diameter:

88% of active picture height (625 line versions);

84% of active picture height (525 line versions).

3.8.2 Color signals outside the circle

The signals are described as vectors, chroma amplitude approx. 330mV (310mV in NTSC version). The component versions of PM 5644 generates signals that will result in these vectors after coding.

From left to right:

Vertical bar with non-alternating R-Y signals (270°/270°) and B-Y = 0 ("Anti-PAL").

The signals are at some points starting at the extreme left of the picture for burst gate check.

2 vertical bars with R-Y signals (270° and 90°) and B-Y = 0.

2 rectangles with signal G-Y = 0 (326° and 146°).

2 rectangles with signal G-Y = 0 (326° and 146°).

2 vertical bars with B-Y signals (0° and 180°) and R-Y = 0.

Vertical bar with line-alternating B-Y signals (0°/180°) and R-Y = 0 ("Anti-PAL").

3.8.3 Background signals

Cross-hatch.

14 horizontal x 18 vertical lines.

Width of vertical lines:

225ns ±5% (D, G, I and L versions)

280ns ±5ns (M, N and P versions).

Background level:

48% of white (D, G, I, L and N version)

52% of white (M and P versions).

Black/white border castellations.**3.9 Pattern composition,
FuBK pattern****3.9.1 Signals in the middle area**

From top to bottom:

Color Bar (including white and black).

EBU (100/0/75/0).

4-riser staircase.**Center cross.****White/black and black/white transistions.****Rise time:**

200ns $\pm$ 5ns.

Multiburst.

100% white reference, 1MHz, 2MHz, 3MHz,
4.43MHz (f_{subc} , G-Y = 0) and 50% white.

Amplitude:

100% of white level.

White bar with black triangle.**Max duration of pulse:**

1 μ s.

Rise time of pulse:

200ns $\pm$ 5ns.

 f_{SUBC} sawtooth.

$\pm$ (R-Y) starting with 75% of white amplitude
(5.4 μ s), then descending to 0% amplitude (16.2 μ s).

+(B-Y) starting with 75% of white amplitude
(5.4 μ s), then descending to 0% amplitude (16.2 μ s).

Anti-PAL signals.

Non-alternating R-Y signals and line-alternating
B-Y signals.

Amplitude:

75% of white level.

Circle.**Diameter:**

93% of active picture height.

3.9.2 Signals outside the middle area**Cross-hatch.**

15 horizontal x 19 vertical lines.

Width of vertical lines:

200ns $\pm$ 5%.

Background level:

25% of white.

3.10 Text insertion (PM 8546)**3.10.1 Station identification**

One or two lines of text may be inserted into the picture.

Text font:

Helvetica condensed (proportional).

Number of letters (Philips pattern):

approx. 9 capital letters (upper text field)

approx. 13 capital letters (lower text field).

More characters may be used depending on the type of text - contact Philips for more information.

3.10.2 Clock and calendar

Date and time can be inserted in the picture.

Text font:

Helvetica condensed (fixed spacing).

Date format:

DD-MM-YY, MM-DD-YY or YY-MM-DD.

Time format:

HH:MM:SS.

Stability:

When the instrument is "ON", the clock circuits are locked to the sync pulses of the video signal.

When the instrument is "OFF", the clock circuits are locked to an internal X-tal.

The clock can also be locked to a studio clock (EBU/SMPTE time code or 1Hz synchronizing signal).

3.11 Logo (optional)

A multi colored, high resolution station logo can be inserted in the picture with option PM 8546.

Contact Philips for more information of the design of the logo.

3.12 Synchronization

Following modes for external synchronization are available:

- locking to a black burst or composite video signal.
- locking to a sync signal

Internal selection between subcarrier free running and subcarrier locked to line giving correct line-to-subcarrier phase.

3.12.1 Internal mode

Subcarrier frequency:

4,433,618.75Hz (D, G, and I versions).

4,406,250/4,250,000Hz $\pm 2,000$ Hz (L versions).

3,579,545Hz (M version).

3,575,611.49Hz (P version).

3,582,056.25Hz (N version).

Color subcarrier temperature stability:

0-50°C : better than 1ppm
(ref. 25°C).

Ageing : less than 0.1ppm
per month.

Stability Sc-H:

Jitter : less than $\pm 2^\circ$.

Drift : less than $\pm 2^\circ$ per 10°C.

Absolute phase : $0^\circ \pm 15^\circ$ line 1 field 1.

3.12.2 External mode

SLOW LOCK:

Requirements as described under EXTERNAL SYNC INPUT.

SYNC LOCK:

Horizontal frequency lock range:

± 1 ppm.

Lock-in time (vertical):

less than 7s .

Jitter with respect to input sync:

less than 6ns for noise free signal of nominal amplitude and frequency.

Jitter for 100% hum (max. 1V_{PP}):

less than 25ns.

Jitter for 28dB_{RMS} signal-to-noise:

less than 10ns.

Line phase change for sync level nominal ± 6 dB:

less than 15ns.

Line phase adjustment:

$\pm 3\mu$ s via front panel potentiometer.

SUBCARRIER LOCK:

Lock range:

± 25 Hz of nominal.

Lock-in time:

less than 1s.

Jitter with respect to incoming burst phase:

less than 1° .

Subcarrier phase adjustment:

more than 360° via front panel potentiometer.

In case no burst is available two lock modes are internally selectable by jumper setting.

– free-running subcarrier

– subcarrier is locked to line frequency with correct

Sc-H phase:

Subcarrier jitter : less than 2° .

Lock range : ± 5 ppm.

3.13 Environmental conditions

The environmental data mentioned in this instruction manual is based on the results of the manufactures procedures. Details of these procedures and failure criteria are supplied on request by the Philips organization in your country, or by PHILIPS, INDUSTRIAL & ELECTRO-ACOUSTIC DIVISION, EINDHOVEN, THE NETHERLANDS.

3.13.1 Temperature range

Operating range:

+5°C to +45°C (+141°F to +113°F)

Functioning temperature:

0°C to +55°C (+32°F to 131°F)

Storage and transport:

-40°C to +70°C (-40°F to +158°F)

3.13.2 Mechanical test conditions

VIBRATION

Limit range for storage and transport:

30min. in each of three directions, 10 to 150Hz;

0.7mmpp and 50m/s² max. acceleration.

According to IEC-Publ. 68, test Fc.

NOTE: Unit mounted on vibration table without shock absorbing material.

BUMP

Limit range for storage and transport:

1000 bumps of 100m/s² sine, 6ms duration in each of 3 directions. According to IEC-Publ.68, test Eb.

PACKAGING

According to UN-D-1400

The test methods mentioned in the N.V. Philips standard UN-D-1400 are in accordance with those of relevant ISO-standards.

3.14 Mains supply conditions

Electrical safety:

according to IEC 348.

Mains supply voltage:

100, 120, 220 or 240V AC (max. 250V), +10%/-15%.

Mains supply frequency:

48-65Hz.

Power consumption:

25W.

3.15 Mechanical data

19" table/rack mount cabinet.

Height	: 44mm	(1.73").
Width	: 483mm	(19").
Depth	: 461mm	(18.14").
Weight	: 6.5kg	(14.32lbs).

4. Accessories & options

4.1 Accessories

Item	Quantity	Ordering number
Mains cable	1 ¹	5322 321 10338
Mains cable (US)	1 ¹	5322 321 10123
Instruction manual	1	9499 493 07211

4.2 Options

Item	Ordering number
------	-----------------

Text and clock generator, 625 lines, PM 8546G/00	9449 085 46003
---	----------------

Text and clock generator, 625 lines, component version, PM 8546G/10	9449 085 46013
--	----------------

Text and clock generator, 525 lines, PM 8546M/00	9449 085 46008
---	----------------

Text and clock generator, SECAM, PM 8546L/00	9449 085 46007
---	----------------

RS232 Remote Control Interface, PM 9190

IEEE Remote Control interface, PM 9191

1: Mains cable is only delivered for the country specified by your order.

INSTALLATION AND OPERATING INSTRUCTIONS

5. Installation

5.1 Initial inspection

Check the contents of the shipment for completeness and possible transport damage. If the contents are incomplete or damaged, a claim should be filed with the carrier immediately, and the Philips Sales or Service organization should be notified in order to facilitate the repair or replacement of the instrument.

5.2 Safety instruction

5.2.1 Earthing

Before any other connection is made, the instrument must be connected to a protective earth conductor via the three-core mains cable.

Before connecting the equipment to the mains of the building installation, the proper functioning of the protective earth lead of the building installation needs to be verified.

WARNING: Any interruption of the protective conductor inside or outside the instrument, or disconnection of the protective earth terminal, is likely to make the instrument dangerous. Intentional interruption is prohibited.

5.3 Mains voltage setting and fuses

Before inserting the mains plug into the mains socket make sure that the instrument is set to the local mains voltage.

NOTE: If the mains plug has to be adapted to the local situation it should be done by a qualified person only.

The mains connector of this instrument is specially constructed for safety. It is impossible to change the mains fuse or alter the operating voltage without removing the mains cable first.

The instrument can be used with a mains voltage supply of 100V, 120V, 220V or 240V (max.250V) +10%-15%.

- Remove the mains cable.
- Slide the protective plastic cover to the left to expose the fuse and coded mains plug. See fig. 5-1.

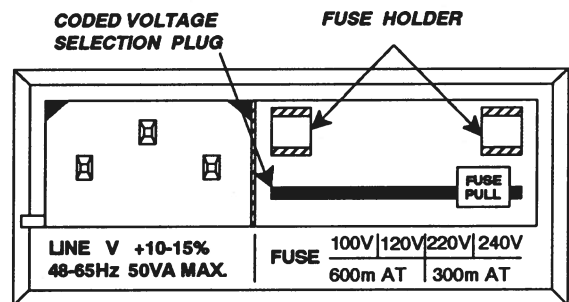


Fig. 5-1 Mains connector.

- Remove the fuse by pulling the black lever.
- The text printed on the coded mains plug is now visible.
- If the text does not agree with the local mains supply, insert a pointed instrument into the hole and remove the coded plug.
- Re-insert the plug so that the visible text agrees with the local mains voltage. (This is the number on the top-left portion of the card).
- Insert the mains fuse with one of the proper rating.
- Slide the protective cover to right.
- Re-insert the mains cable.

WARNING: If mains voltage has to be adapted to the local supply, it must only be done by a qualified person who is aware of the hazards involved. Make sure that only fuses of the required current rating and specified type are used for renewal. The use of repaired (jumpered) fuses and/or the short circuiting of the fuse holders is prohibited. Fuses must only be replaced by a qualified person who is aware of the hazards involved.

5.4 Rack mounting

This Philips PTV instrument is delivered in a 19" cabinet. In systems where several cabinets are mounted in a 19" rack, special attention must be paid to the temperature. To avoid overheating, we recommend the following solutions:

5.4.1 Free air convection

Mount an air-flow unit (e.g. type PM 9799) between or underneath the cabinets. The dimensions of this unit is 1U high (=4.5cm/13/4") and 19" wide.

5.4.2 Forced circulation

Mount a ventilation unit (e.g. PE1373 mounted with 2 fans PE1374) between or underneath the cabinets. The dimensions of this unit are 1U high (=4.5cm/13/4") and 19" wide.

5.5 Access to and replacement of parts

5.5.1 Safety

The opening of covers or removal of parts, except those to which access can be gained by hand, is liable to expose live parts. Accessible terminals may also be live.

The instrument must be disconnected from all voltage sources before performing any adjustment, replacement, maintenance, or repair which requires the instrument to be opened. If adjustment, maintenance, or repair of the opened instrument is unavoidable, it must only be carried out by a skilled person who is aware by hazards involved.

5.5.2 Access to units 1, 2, 3 and 4.

- If the instrument is rack-mounted, remove it from the rack.
- Remove the two rear panel protecting brackets by loosening the screws "B" (see fig. 5-4).
- Remove the top-cover by sliding it towards the rear of the instrument.
- All units of the instrument are now accessible (see fig. 5-2).
- To obtain full access to the main board unit 1, remove the screws securing the text and clock generator unit 4 (where applicable), and flip unit 4 towards the front of the instrument.
- If access to the solderside of the PCB's (unit 1, 2 and 3) is necessary, remove the bottom cover by sliding it towards the rear of the instrument.

5.5.3 Removal of units 2, 3 and 4.

- Get access to the units as described in paragraph 5.5.2.
- Loosen the screws "A" (see fig. 5-3) and remove the instruments front panel.
- Remove the screws "C" securing the unit(s) of interest and pull the unit(s) forward (see fig. 5-5).

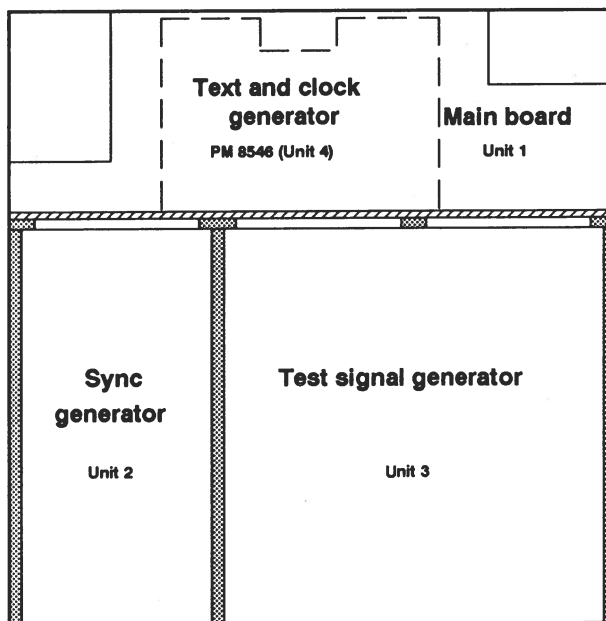


Fig. 5-2 Unit location, top view

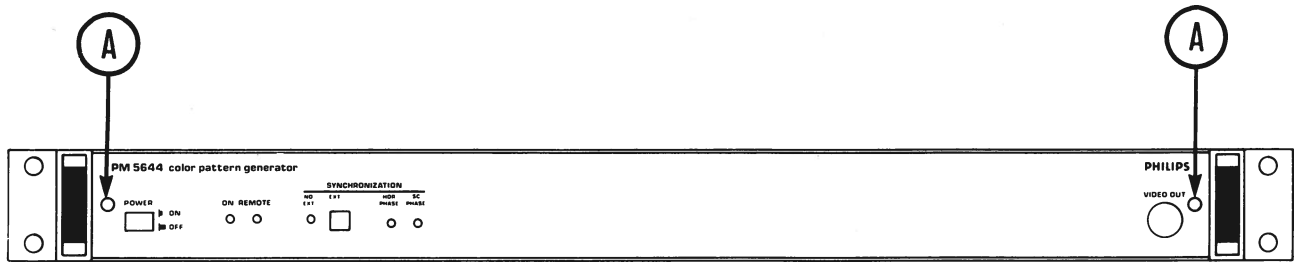


Fig. 5-3 Removal of the front panel

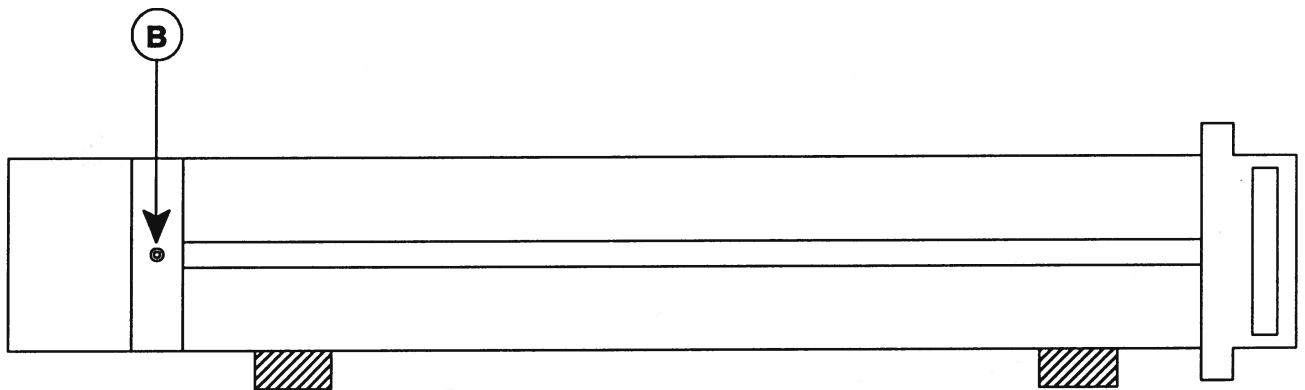


Fig. 5-4 Removal of the rear brackets

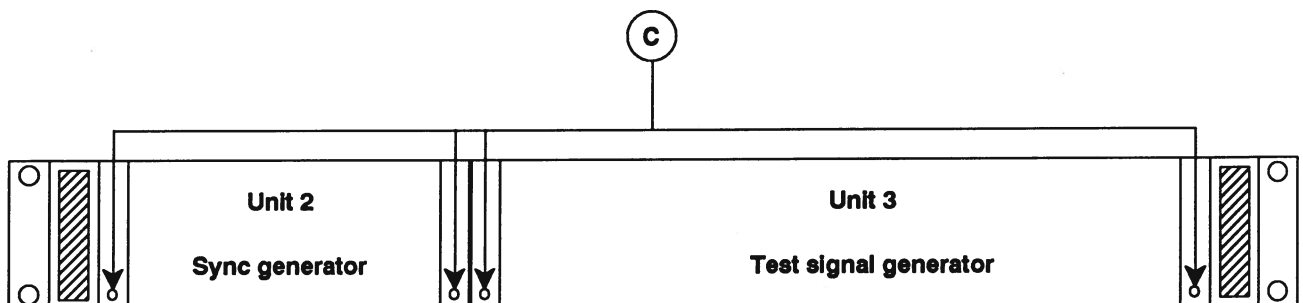


Fig. 5-5 Unit location, front

6. Configuration

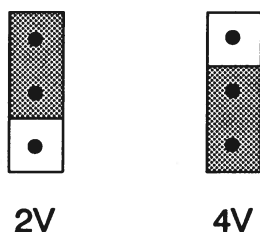
6.1 General

At delivery the instrument is set-up for the version ordered. Modification from one version to the other requires operations that are outside the scope of this manual. This chapter deals with the instrument functions that can be re-configured without the need for special spares and specialized test equipment. However, the correct settings of the version setting programming plugs are also shown below in order to enable verification of the version set-up in a repair situation. For the location of the relevant programming plugs (PP), *please refer to figure 6-3* for PAL and NTSC versions, except when otherwise noted.

6.2 Configuration of the output signals

6.2.1 Sync level

The pulse amplitude on SYNC OUT can be programmed to either 2V_{PP} or 4V_{PP} by positioning PP3 in unit 1 as follows:



↑
Rear of the
instrument

6.2.2 Color ID

The color ID pulse in the black burst line 7, field 1 (line 11, field 1 for NTSC) can be switched on/off by positioning PP5 in unit 1 as follows:

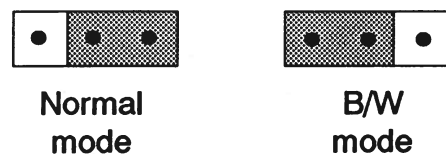


6.3 Synchronization

6.3.1. Gen-lock mode

When the instrument is operated in gen-lock mode (EXT SYNC) it is normally required that both line sync and burst information is supplied by the external SPG (the PM 5644 sync generator is then locked to the external line sync reference and the internal subcarrier generator is locked to the external burst). However, by rearranging the position of jumper PP2 in unit 1 it becomes possible to gen-lock the instrument to a black and white signal (sync without burst). To maintain correct SC/H relationship, the internal sync generator is then locked to the external sync reference while the internal subcarrier generator is locked to the internal line sync generator via the subcarrier coupler facility provided by the sync generator.

To select between normal gen-locking and the special B/W mode, position PP2 in unit 1 as follows:



6.4 Text and logo configuration

The optional text and clock generator PM 8546 can be user configured with regard to a number of functions. The configurations are carried out partly by means of the special function buttons located behind the front panel and partly by means of programming plugs located on PM 8546.

6.4.1 Date format

The 'date format' of the PM 8546 clock can be set to either YY-MM-DD, DD-MM-YY or MM-DD-YY by proceeding as follows:

- Switch off the instrument.
- Remove the front panel and the top cover.
- In the test signal generator unit 3, move the programming plug PP5 from its neutral parking position to PP2, special functions. (see fig. 6-3).
- Switch on the instrument.
- Press S1 to select 'date format' mode (see fig. 6-1).
- Press the status switch S20 (**EXT**) until the desired format is indicated by S8 and S9 (see table 6-1).
- Switch off the instrument.
- Return the programming plug from PP2 to its neutral parking position PP5.

Format	S8	S9
YY-MM-DD	○	●
DD-MM-YY	●	○
MM-DD-YY	●	●

Table 6-1 Date format configuration.

6.4.2 Clock windows

When the PM 8546 clock is installed, windows for accommodation of the 'time' and 'date' display may be opened in the test pattern as follows:

- Switch off the instrument.
- Remove the front panel and the top cover.
- In the test signal generator unit 3, move the programming plug PP5 from its neutral parking position to PP2, special functions. (see fig. 6-3).
- Switch on the instrument.

- Press S2 to select 'Clock window' mode (see fig. 6-1).
- Press the status switch S20 (**EXT**) until the desired format is indicated by S8 and S9 (see table 6-2).
- Switch off the instrument.
- Return the programming plug from PP2 to its neutral parking position PP5.

FORMAT	S8	S9
no window	○	●
time window	●	○
time/date window	●	●

Table 6-2 Clock window configuration

Note: Some instrument versions (FuBK) allow only selection of option 1 and 3 ('no window' and 'time/date' window)

6.4.3 Date on/off

The read out of 'date' may be disabled by removing the programming plug from PP1, PM 8546 (for the location of PP1, refer to fig. 6-2)

6.4.4 Selection of text and logo

The lay-out of the text and logo is to be defined at the time of ordering and is then programmed at the factory. Dependent on the complexity of the defined text and logo (occupied memory space), up to four different lay-outs may be selectable by means of PP2 and PP3, PM 8546 (please refer to table 6-3 and fig. 6-2).

Text/Logo	PP2	PP3
No.1		
No.2		
No.3		
No.4		

Table 6-3 Logo and text selection

6.4.5 Time input format

The PM 8546 can be configured to accept either EBU/SMPTE timecode or 1Hz synchronization by means of PP6 and PP7 (please refer to table 6-4 and fig. 6-2)

Input format	PP6	PP7
Time code		
1Hz		

Table 6-4 Time sync format

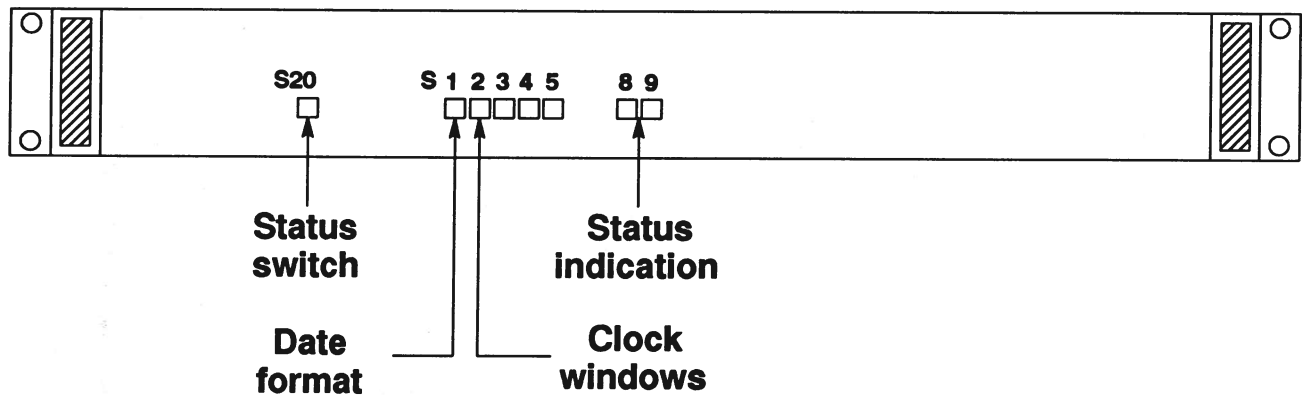


Fig. 6-1 Special configuration, front panel removed

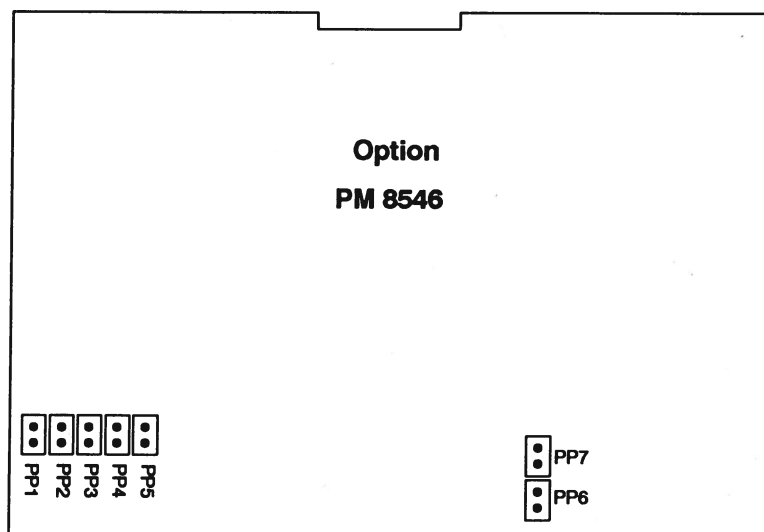


Fig. 6-2 Configuration, text and clock generator

6.5 Test mode

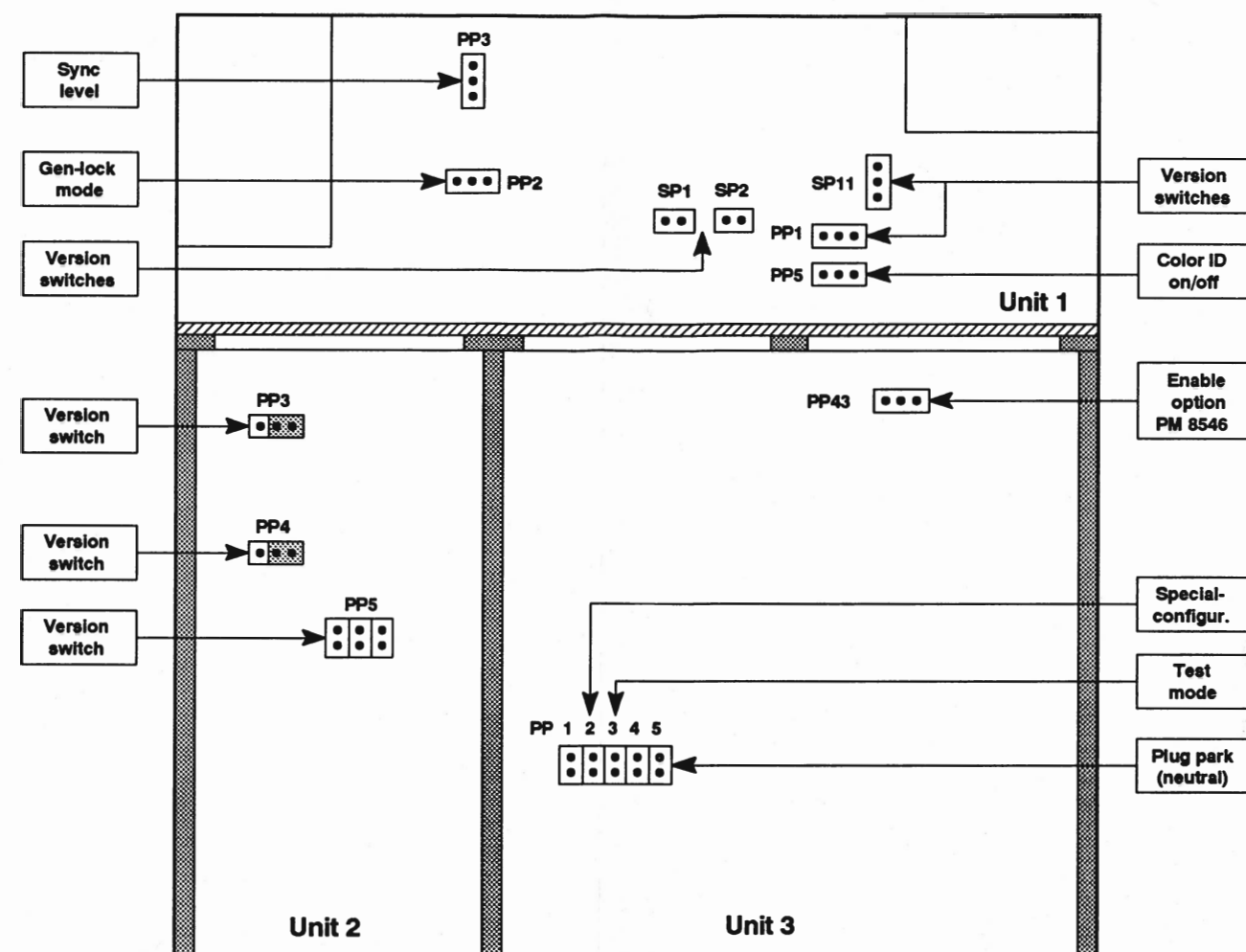
A diagnostic self test of the controller in unit 3 will be activated if a programming plug is present on PP3 in unit 3 during power up of the instrument.
(For details on the self test procedure, please refer to chapter 15).

6.6 Version dependent configuration

This paragraph indicates the correct position of the version defining jumpers located in unit 1 and 2. The provided information serves only as a reference for verification of the plug position of a specific version (it can not be used as a modification guide).

	PAL - B/G, D/K, I	PAL - N	PAL - M	NTSC
SP1, unit 1				
SP2, unit 1				
SP11, unit 1				
PP1, unit 1				
PP2, unit 2				
PP3, unit 2				
PP4, unit 2				
PP5, unit 2				
PP6, unit 2				
PP7, unit 2				
PP8, unit 2				
PP9, unit 2				
PP11, unit 2				
PP12, unit 2				

↑
Rear of the
instrument

*Fig. 6-3 Location of programming plugs (PP)*

7. Operating instructions

7.1 Controls and connections

Introduction.

To provide a quick answer to questions like: "what does this button do?", the following description has been carried out in a location rather than an operation oriented sequence. For the location of the controls and connections, reference should be made to figures 7-2 and 7-3.

An operation oriented description is provided in paragraph 7.2, "Instrument operation".

7.1.1 Front panel controls

SK1 "POWER"

The mains is switched on when this button is pressed, connecting power to the internal DC supply.

LA1 "ON"

A green LED that is lit when DC power is available from the internal DC supply.

LA2 "REMOTE"

A red LED that switches on when the instrument receives a command via the optional remote interface. The LED switches off again (instrument returns to local mode) when a button on the instrument's front panel is pressed.

LA3 "NO EXT"

A red LED that switches on if EXT sync mode has been selected without a sync signal being present on the EXT SYNCR input (the instrument switches automatically to internal sync mode until an external sync signal becomes available).

S20 "EXT"

A switch that when pressed toggles the instrument between internal- and external sync mode. When the instrument is in EXT sync mode the integral LED of the switch is turned on.

P1 "HOR PHASE"

A screwdriver operated control which is active only when the instrument is in genlock mode (EXT sync mode). It allows matching of the phase between

the internally generated line sync pulses and the sync pulses of the external master SPG.

P2 "SC PHASE"

A screwdriver operated control which is active only when the instrument is in genlock mode (EXT sync mode). It allows matching of the phase between the internally generated burst signal and the burst of the external master SPG.

BU1 "VIDEO OUT"

A 75Ω BNC connector on which the generated test pattern is available.

7.1.2 Rear panel connections

BU2 "REMOTE"

Connector for remote control of the instrument (optional). The remote control facility can be supplied for the IEEE448 and RS232 standard. For details please refer to paragraph 7.3., "Remote control".

BU3 "BL B OUT"

A 75Ω BNC connector providing a black burst output (nominal level).

BU4 "VIDEO OUT"

A 75Ω BNC connector on which the generated test pattern is available.

BU5 "VIDEO OUT"

A 75Ω BNC connector on which the generated test pattern is available.

BU7 "TIME CODE IN"

An XLR connector that can be configured for input of either EBU/SMPTE time code or 1Hz synchronization (see paragraph 6.6.3).

Pin config., EBU/SMPTE:

Pin	Function
1	GND
2	Signal
3	Signal

Pin config., 1Hz:

Pin	Function
1	GND
2	GND
3	Signal IN

BU8 "AUDIO OUT"

A 600 Ω XLR connector providing a 1kHz/0dBm audio signal.

BU9 "SYNC OUT"

A 75 Ω BNC connector providing composite sync (amplitude either 4V_{PP} or 2V_{PP} dependent on configuration, *see paragraph 6.2.1*).

BU10-11 "EXT SYNCHR"

Two BNC sockets providing a looped through input facility for an external reference (nominal video level).

BU12 Mains connection

The mains power connector with integrated fuse holder and voltage selector (for selection of voltage and change of fuse, *see paragraph 5.3*).

7.1.3 Special controls

Behind the front panel a number of special function buttons are located. The buttons are assigned different functions dependent on whether the instrument is in normal operation, in configuration mode or in test mode. During normal instrument operation (no programming plug mounted on PP2 and PP3 in unit 3), the functions are as described below (*refer to fig. 7-1*). The functions during configuration- and test mode are described in chapter 6 and 13 respectively.

S1 DATE

A switch that when pressed toggles the DATE set facility on/off.

The integral LED of S1 is lit when DATE set mode is active.

S2-S4

Three switches for setting either 'Hour, Minute, Second' or 'Day, Month, Year' dependent on whether the clock set - or the date set mode has been activated (*see also table 7-1*).

S5 TIME

A switch that when pressed toggles the TIME set facility on/off.

The integral LED of S5 is lit when TIME set mode is active.

S8 NORMAL

When the service signals available via S9 have been used, pressing S8 will restart the generation of the composite test pattern.

S9 TEST

A button that when pressed will activate various test signals suitable for maintenance and service work on the instrument (these test signals will replace the composite test pattern on all video outputs!)

7.2 Instrument operation

7.2.1 Gen-locking

If it is desired to gen-lock PM 5644 to an external master SPG please proceed as follows:

- Connect the external SPG to BU10 "EXT SYNCHR" at the rear panel. If unused, terminate the loop-through BU11 with 75 Ω .
- Press S20 **EXT** on the frontpanel.
- The phase between the master SPG and PM 5644 can now be matched by means of P1 "HOR PHASE" and P2 "SC PHASE" located on the front panel.

7.2.2 Setting of time and date

If the digital clock facility available with the option PM 8546 is not driven by a timecode signal (EBU/SMPTE), it must be set manually by means of five switches S1-S5 located behind the front panel (*see fig 7-1 and table 7-1*).

Setting of date:

- Verify that the clock is set up for your preferred date format (*see paragraph 6.4.1*).
- Press S1 **DATE** to enable the date setting switches S2-S4.